

Администрирование локальных сетей

Лабораторная работа №8

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Цели и задачи работы

Приобретение практических навыков по настройке сетевых сервисов DNS и DHCP в локальной сети, а также проверка автоматического получения IP-адресов клиентскими устройствами.

Ход выполнения работы

Добавление и подключение DNS-сервера

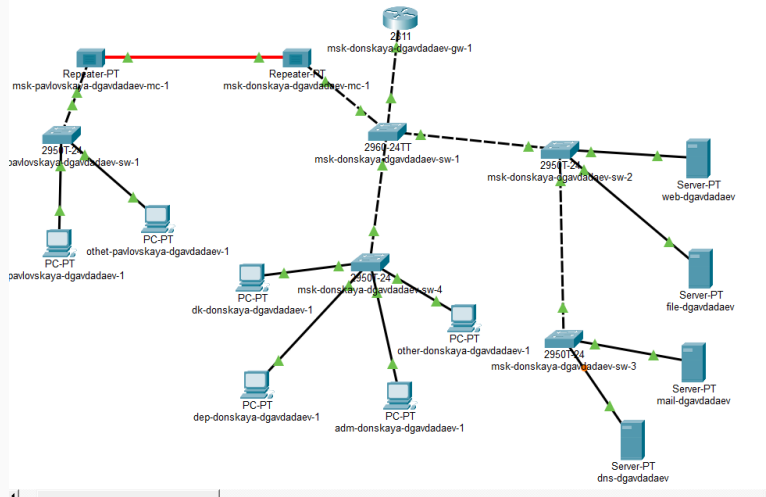


Рис. 1: Топология сети с DNS-сервером

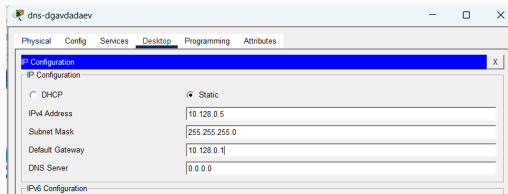


Рис. 2: Настройка IP DNS-сервера

Настройка DNS-сервера

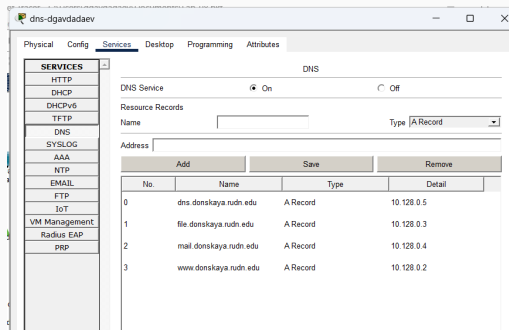
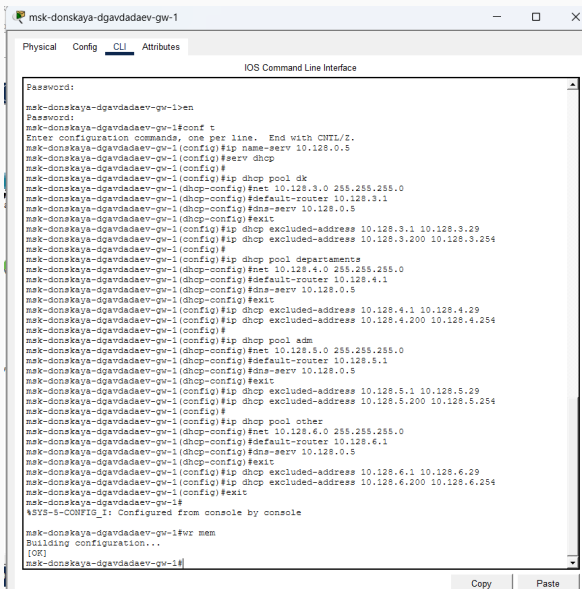


Рис. 3: Настройка службы DNS

Настройка DHCP на маршрутизаторе



```
msk-donskaya-dgavdadaev-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

Password:
msk-donskaya-dgavdadaev-gw-1>en
Password:
msk-donskaya-dgavdadaev-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-dgavdadaev-gw-1 (config)#ip name-serv 10.128.0.5
msk-donskaya-dgavdadaev-gw-1 (config)#serv dhcp
msk-donskaya-dgavdadaev-gw-1 (config)#
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp pool dk
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#net 10.128.3.0 255.255.255.0
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#default-router 10.128.3.1
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#dns-serv 10.128.0.5
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#exit
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp excluded-address 10.128.3.200 10.128.3.254
msk-donskaya-dgavdadaev-gw-1 (config)#
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp pool departaments
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#net 10.128.4.0 255.255.255.0
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#default-router 10.128.4.1
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#dns-serv 10.128.0.5
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#exit
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-dgavdadaev-gw-1 (config)#
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp pool adm
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#net 10.128.5.0 255.255.255.0
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#default-router 10.128.5.1
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#dns-serv 10.128.0.5
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#exit
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
msk-donskaya-dgavdadaev-gw-1 (config)#
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp pool other
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#net 10.128.6.0 255.255.255.0
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#default-router 10.128.6.1
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#dns-serv 10.128.0.5
msk-donskaya-dgavdadaev-gw-1 (dhcp-config)#exit
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp excluded-address 10.128.6.1 10.128.6.29
msk-donskaya-dgavdadaev-gw-1 (config)#ip dhcp excluded-address 10.128.6.200 10.128.6.254
msk-donskaya-dgavdadaev-gw-1 (config)#exit
msk-donskaya-dgavdadaev-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dgavdadaev-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-dgavdadaev-gw-1#
```

Copy Paste


```
msk-donskaya-dgavdadaev-gw-1#sh ip dhcp pool

Pool dk :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 2
  Excluded addresses : 8
  Pending event : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  10.128.3.1         10.128.3.1 - 10.128.3.254    2 / 8 / 254

Pool departments :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Excluded addresses : 8
  Pending event : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  10.128.4.1         10.128.4.1 - 10.128.4.254    1 / 8 / 254

Pool adm :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Excluded addresses : 8
  Pending event : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  10.128.5.1         10.128.5.1 - 10.128.5.254    1 / 8 / 254

Pool other :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 2
  Excluded addresses : 8
  Pending event : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  10.128.6.1         10.128.6.1 - 10.128.6.254    2 / 8 / 254
msk-donskaya-dgavdadaev-gw-1#
```

Рис. 5: Информация о пулах DHCP

```
msk-donskaya-dgavdadaev-gw-1#  
msk-donskaya-dgavdadaev-gw-1#sh ip dhcp binding  
IP address      Client-ID/  
                Hardware address      Lease expiration      Type  
10.128.3.30      0060.2F01.4D20      --                    Automatic  
10.128.3.31      00E0.A3B7.21C3      --                    Automatic  
10.128.4.30      0060.472E.99C       --                    Automatic  
10.128.5.30      00D0.D34B.096A      --                    Automatic  
10.128.6.30      0001.6369.905B      --                    Automatic  
10.128.6.31      0001.97C2.5277      --                    Automatic  
msk-donskaya-dgavdadaev-gw-1#
```

Рис. 6: Таблица выданных адресов

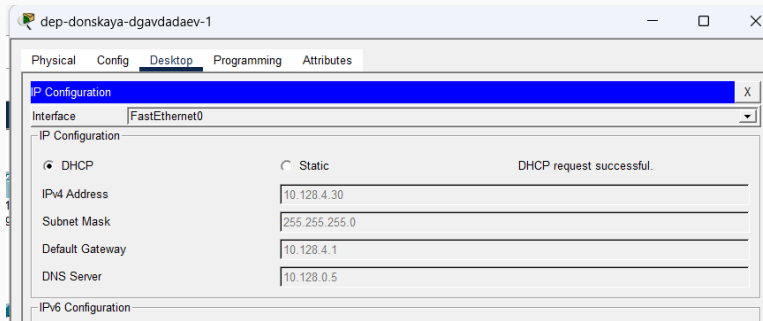
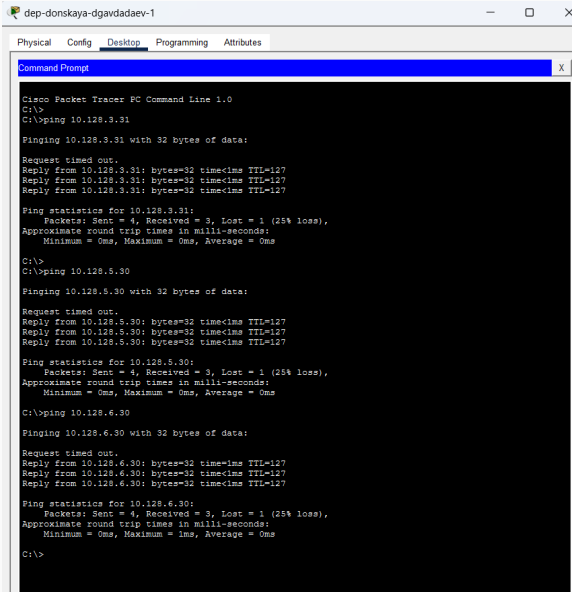


Рис. 7: Получение IP-адреса по DHCP

Проверка связности сети



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'dep-donskaya-dgavdadaev-1'. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with 'Desktop' selected. The Command Prompt shows the following commands and output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.128.3.31

Pinging 10.128.3.31 with 32 bytes of data:

Request timed out.
Reply from 10.128.3.31: bytes=32 time<1ms TTL=127
Reply from 10.128.3.31: bytes=32 time<1ms TTL=127
Reply from 10.128.3.31: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.31:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>ping 10.128.5.30

Pinging 10.128.5.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.5.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.6.30

Pinging 10.128.6.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.6.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

DHCP Discover

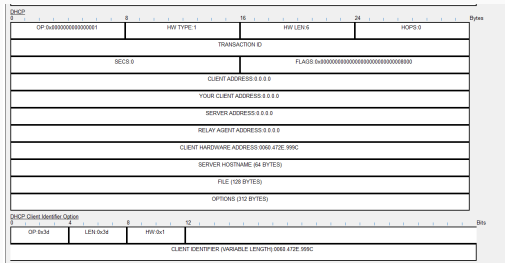


Рис. 9: DHCP Discover

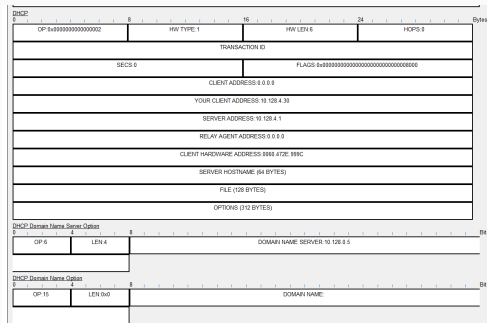


Рис. 10: DHCP Offer

DHCP Request

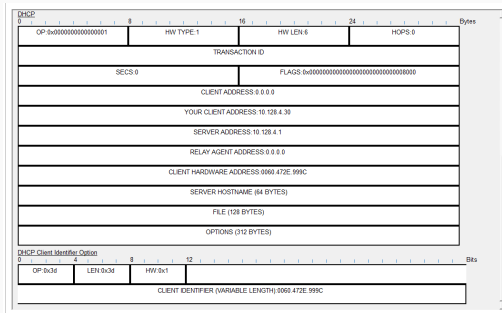


Рис. 11: DHCP Request

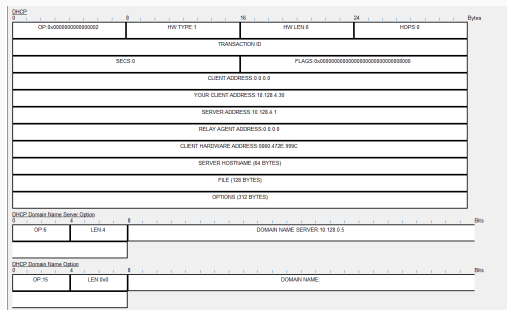


Рис. 12: DHCP ACK

Выводы

В ходе выполнения лабораторной работы:

- добавлен и настроен DNS-сервер
- настроена служба DHCP на маршрутизаторе
- созданы пулы адресов для нескольких подсетей
- клиентские устройства переведены на динамическое получение адресов
- подтверждена корректная работа DHCP и сетевой связности
- изучен процесс обмена DHCP-сообщениями в режиме симуляции